MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Skill Enhancement Course (SEC) - MG4SECAQC200 - AQUAFARM MANAGEMENT
(2024 ADMISSION ONWARDS)
Duration: 1 Hour 30 Minutes, Maximum Marks: 50
Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer questions
Answer all questions
Each question carries **1** marks

1. Name the fatty acids obtained from fishes

Omega 3 fatty acids

2. Which fertilizer is a nitrogen source?

NPK fertilizer, Urea, Cow dung, Poultry excreta (any one of these)

3. What is the purpose of correct way of fish handling?

Reduces stress in fish, Prevents injuries, Prevents diseases and infections, Improves survival rate, Maintains product quality , Ensures safe transport and stocking (any one point)

4. What temperature range is ideal during closed transport of fish?

20°C to 25°C

5. Write an example for protein source ingredient in fish feed

Fish meal, prawn meal, clam meal, bone meal, slaughter house waste, groundnut oil cake, coconut oil cake, sunflower oil cake, gingelly oil cake (any one)

6. What are unwanted plants in an aquaculture pond called?

Aquatic weeds

7. Why is biofencing considered eco-friendly?

Biofencing is considered eco-friendly because it uses living plants or shrubs as a fence instead of artificial materials like metal, plastic, or concrete.It improves biodiversity, reduce pollution, enhances the environment

8. Why fish are released into the pond only after quarantine?

Fish are quarantined before release to detect diseases, treat infected fish, protect the health of fish already present in the pond and also for acclimatization purpose

9. Which mineral supports fish bone formation?

Calcium

10. Write one method of mechanical aeration

Paddle wheel aerator, Propeller aerator, Diffused air aerator (air blower), Spray aerator , Pump aerator

Part B

Short Answer Questions

Answer any 10 questions

Each question carries 2 marks

11. How do raceways differ from ponds?

Raceways differ from ponds in that raceways are long, narrow tanks with a continuous flow of water, which provides better oxygen supply and waste removal for fish. Ponds are larger, still water bodies where water exchange is limited and fish are cultured in a more natural environment.

12. What is the role of climate in aquaculture facility planning?

Climate plays an important role in aquaculture facility planning because factors such as temperature, rainfall, wind and sunlight affect water quality and fish growth. Proper understanding of climate helps in selecting suitable sites, designing ponds and ensuring healthy fish production throughout the year.

13. Why should harvested aquatic organisms be handled gently?

Harvested aquatic organisms should be handled gently to avoid physical damage to their body such as bruising, scale loss or injury. Gentle handling also reduces stress and helps maintain the freshness, quality and market value of the harvested organisms.

14. Why is carbohydrate important in fish diets?

Carbohydrates are important in fish diets because they provide a cheap and easily available source of energy for fish. They help spare proteins for growth and also improve the utilization of other nutrients in the feed.

15. Mention any two reasons for concurrence of diseases in aquatic organisms.

Diseases in aquatic organisms occur due to poor water quality and overcrowding in culture systems. They may also be caused by pathogens such as bacteria, viruses, parasites and fungi which infect weakened or stressed organisms.

16. What is meant by sustainable aquaculture?

Sustainable aquaculture means the farming of aquatic organisms in a way that meets present needs without damaging the environment or depleting natural resources. It ensures long-term production while maintaining ecological balance and protecting aquatic ecosystems.

17. What are the common methods used for eradication of weed fishes?

Common methods used for eradication of weed fishes include draining and drying of ponds and the use of fish toxicants such as mahua oil cake or rotenone. These methods help eliminate unwanted fishes before stocking cultured species.

18. Why is availability of seed important in species selection for aquaculture?

Availability of seed is important in species selection for aquaculture because a regular and reliable supply of healthy seed ensures successful stocking and continuous production. Lack of good quality seed can affect growth, survival and overall productivity of the culture system.

19. How soil filling is done during construction of dike?

Soil filling during construction of a dike is done by placing soil in layers and compacting each layer properly to make the structure strong and stable. This method helps prevent erosion, leakage and damage to the dike.

20. What precautions are taken during fish pond preparation? Write any two factors

During fish pond preparation precautions are taken such as removing weeds and unwanted fishes, drying and cleaning the pond bottom, repairing pond dikes and ensuring proper inlet and outlet structures. Liming and fertilization are also done to improve soil and water quality before stocking fish. (any two points)

21. How does value addition improve consumer acceptance?

Value addition improves consumer acceptance by enhancing the quality, taste, appearance and convenience of the product. It also increases shelf life and market value, making the product more attractive and desirable to consumers.

22. How does turbidity influence light penetration and phytoplankton productivity in fish pond?

Turbidity reduces the penetration of sunlight into the pond water. When turbidity is high, less light reaches deeper layers, which decreases photosynthesis and reduces phytoplankton productivity in the fish pond.

Part C
Short Essay Questions
Answer any 4 questions
Each question carries 5 marks

23. Explain the importance of washing and cleaning in fish processing.

Removes dirt, slime, blood and microorganisms, Improves hygiene and sanitation during processing, Reduces chances of microbial contamination and spoilage,

Maintains freshness and quality of fish, Improves appearance and market value of the product, Helps in safe handling and further processing of fish

24. Describe the importance of regular monitoring of fish health.

Early detection of diseases and infections, Observation of abnormal behaviour and stress in fish, Maintenance of good water quality and culture conditions, Prevention of disease spread to other fish, Timely treatment and management practices, Improves survival rate, growth and overall productivity, Reduces economic loss in aquaculture operations

25. Describe the export potential of aquaculture products.

High international demand for seafood products, Major export items include shrimp, prawns, fish and molluscs, Source of foreign exchange earnings for the country, Provides employment in farming, processing and export sectors, Development of value-added products for global markets, Improves economic growth and strengthens the fisheries sector

26. Describe the role of soil type in aquaculture site suitability

Soil determines water retention capacity of ponds, Clayey or clay-loam soils are most suitable for pond construction, Prevents seepage and water loss from ponds, Influences pond stability and strength of dikes, Affects nutrient availability and productivity of the pond, Helps maintain good water quality and supports fish growth

27. Discuss the concept of feeding schedule and its significance in aquaculture management.

Feeding schedule means providing feed to fish at fixed times and in proper quantities according to their growth stage., Ensures regular and balanced nutrition for fish growth, Improves feed utilization and feed conversion efficiency, Prevents overfeeding and reduces feed wastage, Maintains good water quality in the culture system, Promotes healthy growth and higher survival rate, Increases productivity and profitability in aquaculture operations

28. Describe the common types of aquatic insects found in aquaculture ponds.

Common aquatic insects in aquaculture ponds include water beetles, water bugs, dragonfly and damselfly larvae, and mosquito larvae, Some insects act as predators of fish eggs and larvae, Others serve as natural food for fish, They influence the ecological balance of pond ecosystem, High abundance of predatory insects can reduce fish seed survival, Regular monitoring and control are important for healthy fish culture.